



UNIVERSITY OF CAPE TOWN

CSC200SS: Computer Science

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1 Parallel and Concurrent Programming

2 Theory of Computation

2.1 Regular Languages

2.1.1 Finite automata

Definition: Finite automata

A **finite automaton** is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where:

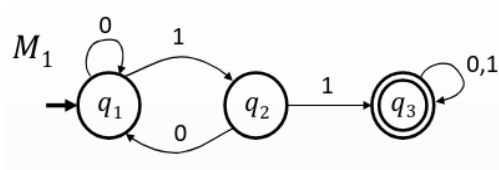
- Q is a finite set of all possible states.
- Σ is a finite set of all alphabet symbols.
- δ is a transition function $\delta : Q \times \Sigma \rightarrow Q$. In other words,

$$\delta(q, a) = r \text{ means } \textcircled{q} \xrightarrow{a} \textcircled{r}$$

- q_0 is the start state.
- F is the set of accept states.

Example:

Let's consider the following finite automaton.



Here, $M_1 = (Q, \Sigma, \delta, q_1, F)$ is a finite automaton. For sake of example, here are the exact details of M_1 .

- $Q = \{q_1, q_2, q_3\}$
- $\Sigma = \{0, 1\}$
- $F = \{q_3\}$

It accepts a finite string as input. The output is either *ACCEPT* or *REJECT*. The computation goes as follows:

1. Begin at start state.
2. Read input symbols.
3. Follow corresponding transitions.
4. *ACCEPT* if we reach the accept state, otherwise *REJECT*.

For example: $01101 \rightarrow \text{ACCEPT}$, while $00101 \rightarrow \text{REJECT}$. Here, M_1 accepts all string in A where $A = \{w | w \text{ contains substring } 11\}$. We say that A is the language of $M_1 \iff M_1$ recognizes $A \iff A = L(M_1)$.

2.1.2 Strings and languages

A **string** is a finite sequence of symbols $w_1w_2\cdots w_n \in \Sigma$. A **language** is a set of string (finite or infinite). The **empty string** ε is the string of length 0. The **empty language** $\emptyset$ is the set with no strings.

Definition: Accepting strings

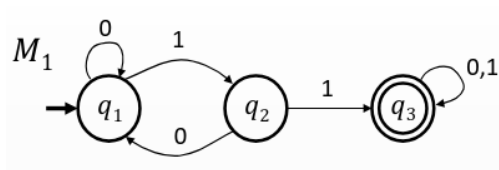
A finite automaton M **accepts string** $w = w_1w_2\cdots w_n$ for each $w_i \in \Sigma$ if there exists a sequence of states $r_0, r_1, \dots, r_n \in Q$ where:

- $r_0 = q_0$
- $r_i = \delta(r_{i-1}, w_i)$ for $1 \leq i \leq n$
- $r_n \in F$

Definition: Regular language

We say a language is **regular** if some finite automaton recognizes it.

Example 1:



$L = (M_1) = \{w | w \text{ contains substring } 11\} = A$. Therefore, A is regular.

Example 2:

Let $B = \{w | w \text{ has an even number of 1s}\}$. In this case, B is regular. (not shown)

Example 3:

Let $C = \{w | w \text{ has equal number of 0s and 1s}\}$. In this case, C is not regular. (provable)

2.1.3 Regular expressions

Definition: Regular operations

Let A, B be languages.

- **Union:** $A \cup B = \{w | w \in A \text{ or } w \in B\}$.
- **Concatenation:** $A \circ B = \{xy | x \in A \text{ and } y \in B\} = AB$.
- **Star:** $A^* = \{x_1 \dots x_k | x_i \in A \text{ for } k \geq 0\}$. (Note: $\varepsilon \in A^*$)

Regular expressions are built from $\Sigma, \emptyset, \varepsilon$ by using regular operations $\cup, \circ, \star$.

Example 1:

- $(0 \cup 1)^* = \Sigma^*$ gives all strings over Σ .
- Σ^*1 gives all strings that end with 1.
- $\Sigma^*11\Sigma^*$ gives all strings that contain 11 = $L(M_1)$.

Example 2:

Let $A = \{\text{good, bad}\}$ and $B = \{\text{boy, girl}\}$.

- $A \cup B = \{\text{good, bad, boy, girl}\}$.
- $A \circ B = AB = \{\text{goodboy, goodgirl, badboy, badgirl}\}$.
- $A^* = \{\varepsilon, \text{good, bad, goodgood, goodbad, badgood, badbad, goodgoodgood, ...}\}$

2.1.4 Nondeterministic finite automata**Definition: Nondeterministic finite automata**

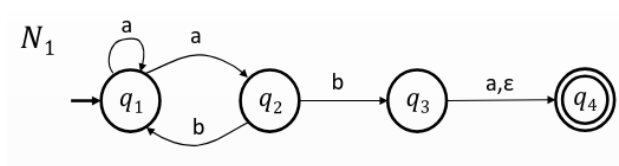
A **nondeterministic finite automata (NFA)** N is a 5-tuple $(Q, \Sigma, \delta, q_0, F)$ where everything is the exact same as a DFA except δ . Here, $\delta : Q \times \Sigma_\varepsilon \rightarrow \mathcal{P}(Q) = \{R \mid R \subseteq Q\}$.

New features of nondeterminism:

- Multiple paths possible (0, 1, or many at each step)
- ε -transition is a "free" move without reading input.
- Accept input if **some** path leads to the accept state.

Nondeterminism doesn't correspond to a physical machine we can build. However, it is useful mathematically. Here are some ways to think about it:

- **Computationally:** Fork new parallel thread and accept if any thread leads to an accept state.
- **Mathematically:** Tree with branches. Accept if any branch leads to an accept state.
- **Magical:** Guess at each nondeterministic step which way to go. The machine always makes the right guess that leads to the accept state, if possible.

Example:

Here, we have $\delta(q_1, a) = \{q_1, q_2\}$, but $\delta(q_1, b) = \emptyset$.

2.1.5 Converting NFAs to DFAs

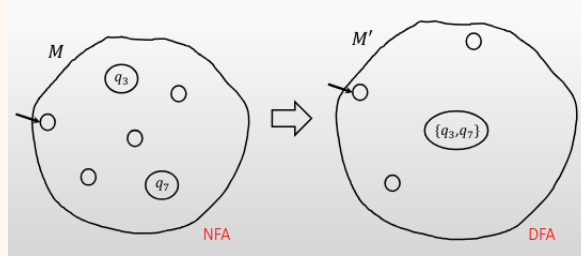
Theorem

If an NFA recognizes A , then A is regular.

Proof:

Let NFA $N = (Q, \Sigma, \delta, q_0, F)$ recognize A . We need to construct a DFA $M = (Q', \Sigma, \delta', q'_0, F')$, ignoring the ε -transitions. Here M is defined as:

- $Q' = \mathcal{P}(Q)$.
- $\delta'(R \in Q', a) = \{q \in Q \mid \delta(r, a) = q \text{ for some } r \in R\}$.
- $q'_0 = \{q_0\}$.
- $F' = \{R \in Q' \mid R \cap F \neq \emptyset\}$.



2.1.6 Closure properties for regular languages

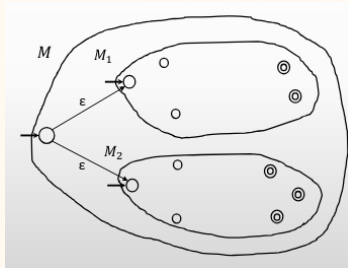
Theorem: Closure under $\cup$.

If A_1 and A_2 are regular languages, then so is $A_1 \cup A_2$.

Proof:

Let $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1 and similarly, let $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize A_2 . We need to construct $M = (Q, \Sigma, \delta, q_0, F)$ where

- $Q = Q_1 \times Q_2 = \{(q_1, q_2) \mid q_1 \in Q_1 \wedge q_2 \in Q_2\}$.
- $q_0 = (q_1, q_2)$.
- $\delta((q, r), a) = (\delta_1(q, a), \delta_2(r, a))$.
- $F = (F_1 \times Q_2) \cup (Q_1 \times F_2)$.

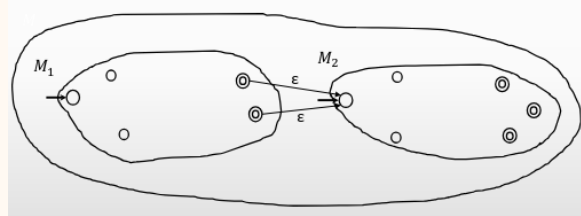


Theorem: Closure under $\circ$.

If A_1 and A_2 are regular languages, then so is A_1A_2 .

Proof:

Let $M_1 = (Q_1, \Sigma, \delta_1, q_1, F_1)$ recognize A_1 and similarly, let $M_2 = (Q_2, \Sigma, \delta_2, q_2, F_2)$ recognize A_2 . We need to construct $M = (Q, \Sigma, \delta, q_0, F)$ where



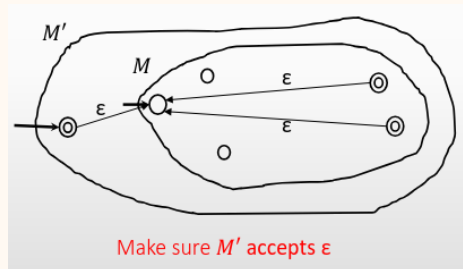
M should accept input w if $w = xy$ where M_1 accepts x and M_2 accepts y . The nondeterministic M has the option to jump to M_2 when M_1 accepts.

Theorem: Closure under $\star$

If A is a regular language, then so is $A^\star$.

Proof:

Given a DFA M recognizing A , construct an NFA M' recognizing $A^\star$.



M' should accept input w if $w = x_1x_2 \dots x_k$ where $k \geq 0$ and M accepts each x_i .

2.1.7 Converting a regular expression to NFA

Theorem

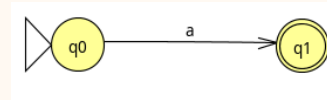
If R is a regular expression and $A = L(R)$, then A is regular.

Proof:

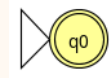
Convert R to the equivalent NFA M .

If R is atomic, an equivalent M is:

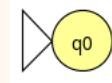
- $R = a$ for $a \in \Sigma$



- $R = \varepsilon$



- $R = \emptyset$



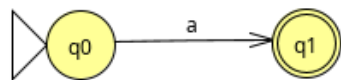
If R is composite:

- $R = R_1 \cup R_2$
- $R = R_1 \circ R_2$
- $R = R_1^*$

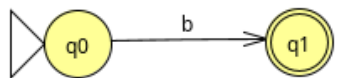
Example:

Convert $(a \cup ab)^*$ to an equivalent NFA.

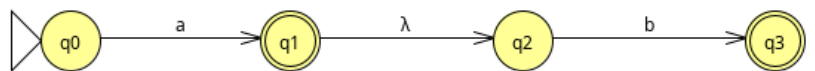
- a :



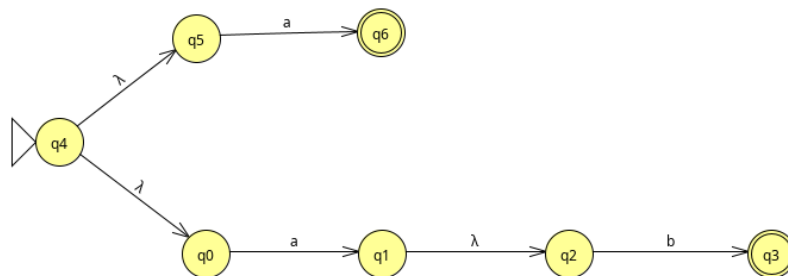
- b :



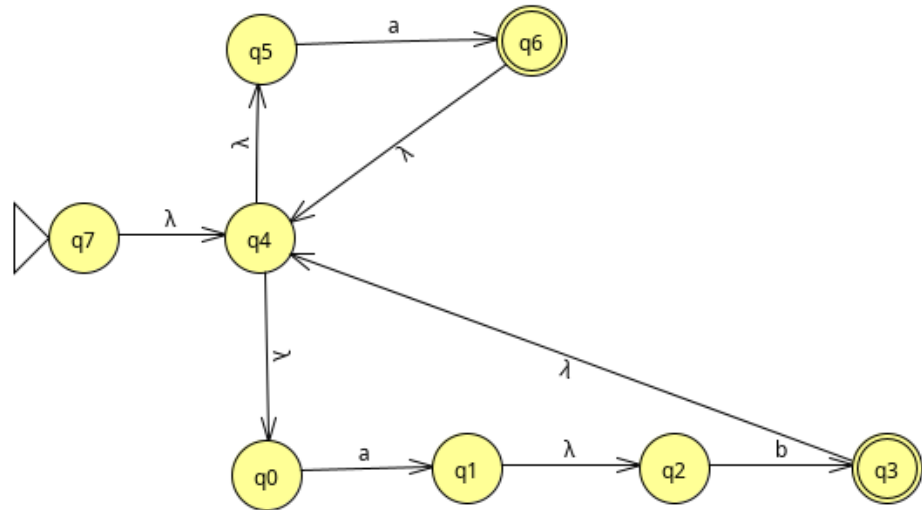
- ab :



- $a \cup ab$:



- $(a \cup ab)^*$:



Reminder

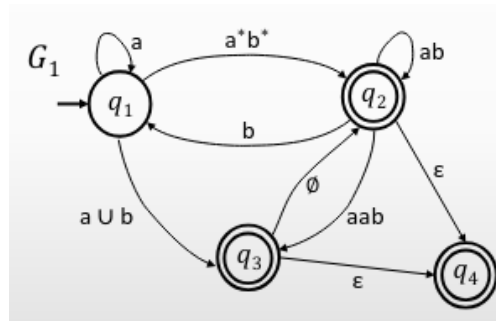
A λ transition is equivalent to an ε transition.

2.1.8 Generalized NFA

Definition: Generalized NFA

A **generalized nondeterministic finite automaton (GNFA)** is similar to an NFA, but allows regular expressions as transition labels.

Example:



For convenience, we will assume:

- One accept state, separate from the start state.
- One arrow from each state to each state, except
 - only exiting the start state
 - only entering the accept state

We can easily modify a GNFA to have this special form.

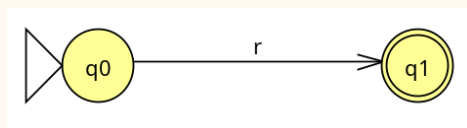
2.1.9 Converting a GNFA to a regular expression

Lemma

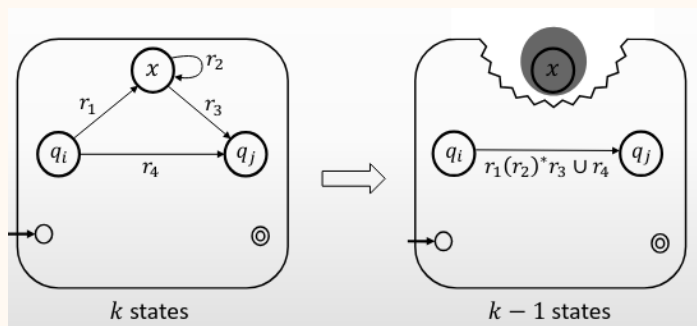
Every GNFA G has an equivalent regular expressions R .

Proof:

We will make use of an inductive argument on the number of states k of G . For the base step ($k = 2$):



Assume the Lemma holds true for $k - 1$ states. Now we want to prove it holds for k states. The central idea is to convert a k -state GNFA to an equivalent $(k - 1)$ -state GNFA.



1. Pick any state x except the start and accept states.
2. Remove x .
3. Repair the damage by recovering all paths that went through x .
4. Make the indicated change for each pair of states q_i, q_j .

Thus DFAs and regular expressions are equivalent. ■

2.1.10 Non-regular languages

How do we show a language is not regular?

- Remember, to show a language is *regular*, we give an equivalent DFA.
- To show a language is *not regular*, we must give a proof.
- It is not enough to say you couldn't find a DFA for it, therefore it isn't regular!

Example 1:

Let $B = \{w | w \text{ has equal numbers of 0s and 1s}\}$. One's intuition says that B is not regular since DFAs cannot count unboundedly.

Example 2:

Let $C = \{w \mid w \text{ has equal numbers of } 01 \text{ and } 10 \text{ substrings}\}$. As examples, $0101 \notin C$ and $0110 \in C$. Again, one's intuition says that C is not regular since DFA cannot count unboundedly. However, C is regular!

2.1.11 Proving non-regularity**Pumping Lemma**

For every regular language A , there exists a number p , called the "pumping length", such that if $s \in A$ and $|s| \geq p$ then $s = xyz$ where

$$1. \ xy^iz \in A \text{ for all } i \geq 0 \quad (y^i = \underbrace{yy \cdots y}_i).$$

$$2. \ y \neq \varepsilon.$$

$$3. \ |xy| \leq p.$$

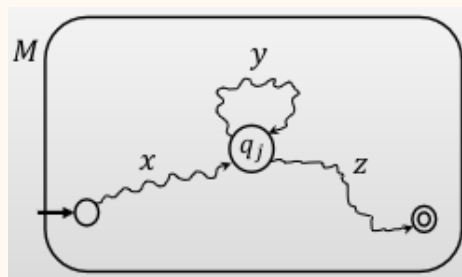
Informally, if A is regular, then every long string in A can be "pumped" and the result stays in A .

Proof:

Let DFA M recognize A . Let p be the number of states in M . Pick $s \in A$ where $|s| \geq p$.

$$s = \begin{array}{c} x \quad y \quad z \\ \hline q_j \quad q_j \end{array}$$

M will repeat a state q_j when reading s because s is so long. The path that M follows when reading s .

**Example 1:**

Let $D = \{0^k 1^k \mid k \geq 0\}$. Show that D is not regular.

Assume, for the sake of contradiction, that D is regular.

Since D is regular, the pumping lemma must hold. Let $s = 0^p 1^p \in D$.

The pumping lemma says that we can divide $s = xyz$.

But $xyyz$ has excess 0s, and thus $xyyz \notin D$ contradicting the pumping lemma.

Therefore, our assumption is false and thus D is not regular.

Example 2:

Let $F = \{ww \mid w \in \Sigma^*\}$. Say $\Sigma^* = \{0, 1\}^*$. Show that F is not regular.

Assume, for the sake of contradiction, that F is regular.

Since F is regular, the pumping lemma must hold.

We need to choose $s \in F$. Let $s = 0^p 10^p 1 \in F$.

We can show that $s = xyz$ cannot be pumped satisfying the 3 conditions.

Thus, $xyyz \notin F$, which is a contradiction. Therefore, F is not regular.

Example 3:

Let $B = \{w \mid w \text{ has equal numbers of 0s and 1s}\}$. Show that B is not regular.

Assume, for the sake of contradiction, that B is regular.

We know 0^*1^* is regular, and so $B \cap 0^*1^*$ is regular (closure under intersection).

But $D = B \cap 0^*1^*$ and we already showed D is not regular (from example 1).

Therefore, B is not regular.

2.2 Context-free languages**2.2.1 Context-free grammars****Definition: Context-free grammar**

A **Context Free Grammar (CFG)** G is a 4-tuple (V, Σ, R, S) where

- V is the finite set of variables.
- Σ is the finite set of terminal symbols.
- R is the finite set of rules (rule form: $V \rightarrow (V \cup \Sigma)^*$).
- S the starting variable.

For $u, v \in (V \cup \Sigma)^*$ write

1. $u \Rightarrow v$ if you can go from u to v with one substitution step in G .
2. $u \xRightarrow{*} v$ if you can go from u to v with some number of substitution steps in G .
 $u \Rightarrow u_1 \Rightarrow u_2 \Rightarrow \cdots \Rightarrow u_k \Rightarrow v$ is called a *derivation* of v from u . Furthermore, if $u = S$ then it is a *derivation* of v .

$L(G) = \{w \mid w \in \Sigma^* \text{ and } S \xRightarrow{*} w\}$.

Definition: Context-free language

A is a **Context Free Language (CFL)** if $A = L(G)$ for some CFG G .

2.2.2 Ambiguity

Take the following two CFGs:

G_1

$E \rightarrow E + T \mid T$

$T \rightarrow T \times F \mid F$

$F \rightarrow (E) \mid a$

G_2

$E \rightarrow E + E \mid E \times E \mid (E) \mid a$

Both G_1 and G_2 recognize the same language, i.e. $L(G_1) = L(G_2)$. However, G_1 is an unambiguous CFG and G_2 is ambiguous.

2.2.3 Pushdown automaton

Definition: Pushdown automaton

A **Pushdown Automaton (PDA)** is a 6-tuple $(Q, \Sigma, \Gamma, \delta, q_0, F)$:

- Σ is the input alphabet.
- Γ is the stack alphabet.
- $\delta : Q \times \Sigma_\varepsilon \times \Gamma_\varepsilon \rightarrow \mathcal{P}(Q \times \Gamma_\varepsilon)$
 $\delta(q, a, c) = \{(r_1, d), (r_2, e)\}$

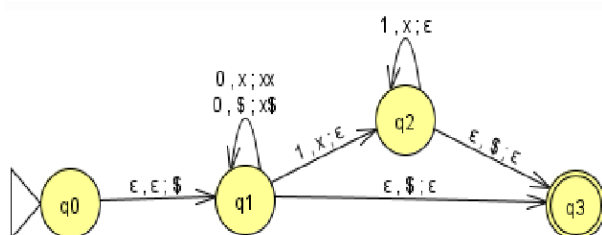
We accept if some thread is in the accept state at the end of the input string.

A PDA operates like an NFA except can write/add (push) or read/remove (pop) symbols from the top of the stack. The input appears on a tape.

Example 1:

PDA for $D = \{0^k 1^k : k \geq 0\}$

1. Read 0s from input, push 0 onto the stack until a 1 is read.
2. Read 1s from input, while popping 0s from the stack.
3. Enter the accept state if the stack is empty.



How to read transitions for PDAs

Example 1: Take the transition $\varepsilon, \varepsilon; \$$.

- The first ε is the input read from the tape. Since we're reading ε , this is equivalent to a blank on the tape.
- The second ε is the value on the top of the stack. Since we're reading ε , we assume the stack is empty.
- The last $\$$ is part of the stack alphabet. This signifies the bottom of the stack. We want to "push" this value onto the stack.

Example 2: Take the transition $\varepsilon, \$; \varepsilon$.

- The first ε is the input read from the tape. Since we're reading ε , this is equivalent to a blank on the tape.
- The $\$$ is the value we expect on the top of the stack.
- The second ε is the value we push onto the stack. We're effectively replacing the bottom-of-stack character with a blank, clearing the stack.

Example 2:

PDA for $B = \{ww^R : w \in \{0,1\}^*\}$ with sample input: 011110.

1. Read and push input symbols onto the stack. Non-deterministically either repeat or go to (2).
2. Read input symbols and pop stack symbols, and compare. If they are ever not equal, then the thread rejects.
3. Enter the accept state if the stack is empty.

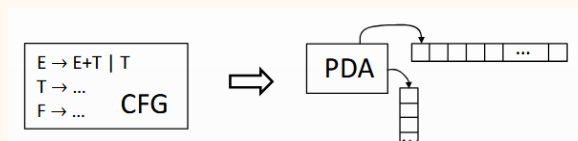
2.2.4 Converting CFGs to PDAs

Theorem

If A is a CFG, then some PDA recognizes A .

Proof:

Convert A 's CFG to a PDA.



1. Push the start symbol onto the stack.
2. If the top of the stack is:
 - *Variable*: replace with right hand side of rule (non-deterministic choice).

- *Terminal*: pop it and match with next input symbol.
3. If the stack is empty, enter the accept state.

2.2.5 Equivalence of CFGs and PDAs

Theorem

A is a CFL if and only if some PDA recognizes A . (*Proof not examinable*)

2.2.6 Chomsky normal form (CNF)

Chomsky normal form is a simplification of grammars:

- If the start symbol is recursive, remove it.
- Removing ε -productions.
- Removing unit productions.
- Removing "useless" productions.

Converting to CNF

Given a grammar $G = \{V, T, S, P\}$, we can derive $G' = \{V, T, S', P'\}$ as follows:

Removing recursive start symbol:

Introduce a new start symbol S' . Let P' be the production rule set P of G with the following rule added to it: $S' \rightarrow S$.